Post-Hoc Compression-Aware Differential Privacy: Optimizing DP Training for Deployed Compressed Models

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Abstract

Differential Privacy (DP) has become the gold standard for protecting training data privacy in machine learning, but DP-SGD suffers from significant accuracy degradation under strong privacy guarantees. This challenge is exacerbated when deploying models on resource-constrained devices, where post-hoc model compression (pruning, quantization) is essential. We identify a critical gap: current DP training is compression-oblivious, allocating privacy budget uniformly regardless of which parameters will survive compression. We propose Post-Hoc Compression-Aware DP-SGD (PHCA-DP-SGD), which incorporates knowledge of anticipated post-hoc compression into the training process through continuous per-parameter clipping weights based on estimated survival probabilities. Unlike pre-pruning approaches that require public data, PHCA-DP-SGD estimates survival probabilities from private training dynamics without additional privacy cost. On CIFAR-10 with ResNet-18, PHCA-DP-SGD achieves 46.73% test accuracy after 70% sparsity compression under $(\epsilon = 3, \delta = 10^{-5})$-DP, compared to 43.25% for standard DP-SGD—a 3.48 percentage point improvement ($p < 0.05$). Our method matches pre-pruning performance without requiring public data, demonstrating that compression-aware training significantly improves privacy-utility trade-offs for deployed models.

1 Introduction

Differential Privacy (DP) dwork2006calibrating has emerged as the gold standard for protecting training data privacy in machine learning, with DP-SGD abadi2016deep being the most widely adopted algorithm for training private deep neural networks. However, DP-SGD suffers from a fundamental privacy-utility trade-off: stronger privacy guarantees (lower ϵ) typically result in significant accuracy degradation. This challenge is particularly acute when deploying models on resource-constrained devices, where model compression via pruning or quantization is essential.

Recent work by li2025compleak revealed a critical but overlooked phenomenon: model compression operations—including pruning, quantization, and weight clustering—exacerbate privacy leakage when adversaries have access to multiple compressed versions of a model. Their experiments demonstrate that compression affects member and non-member samples differently, creating new vulnerabilities that compound existing privacy risks.

This creates a paradox: DP training reduces model capacity through noise injection, then compression further reduces capacity, yet the combination has not been systematically studied for the common deployment scenario where **compression is applied after training** (post-hoc compression). Practitioners currently train with DP-SGD assuming the model will be deployed as-is, then compress it as a separate post-processing step—potentially undermining privacy guarantees in ways that are not accounted for in the privacy budget.

1.1 Key Insight and Contributions

Our key insight is that when post-hoc compression is applied after DP training, the interaction between DP noise and compression-induced information loss creates a complex privacy-utility landscape. Current DP training is compression-oblivious, allocating privacy budget uniformly across all parameters regardless of whether they will survive compression. This is suboptimal: parameters that will be pruned need not receive the same privacy protection investment as those that will be retained.

We propose **Post-Hoc Compression-Aware DP-SGD (PHCA-DP-SGD)**, a training algorithm that jointly optimizes for differential privacy and post-hoc model compression. Unlike pre-pruning approaches adamczewski2023prepruning that require public data for parameter selection, PHCA-DP-SGD uses continuous per-parameter clipping weights based on estimated survival probabilities derived entirely from private training dynamics.

Our contributions are:

- We propose PHCA-DP-SGD, the first method to optimize DP training for post-hoc compression without requiring public data, using continuous per-parameter clipping weights based on estimated survival probabilities.
- We demonstrate that PHCA-DP-SGD achieves 3.48 percentage points higher accuracy than standard DP-SGD after compression (46.73% vs. 43.25%) at $(\epsilon = 3, \delta = 10^{-5})$ on CIFAR-10 with ResNet-18 ($p < 0.05$).
- We show that PHCA-DP-SGD matches pre-pruning baseline performance without using public data, while significantly outperforming importance-based binary masking approaches zhang2025adadpigu.
- We provide a comprehensive ablation study validating that both per-parameter clipping and compression-aware noise calibration contribute to performance gains.

2 Related Work

2.1 Differential Privacy in Deep Learning

dwork2006calibrating introduced differential privacy as a formal privacy framework, and abadi2016deep proposed DP-SGD, which has become the standard algorithm for training private deep neural networks. DP-SGD clips per-sample gradients to bound sensitivity and adds calibrated Gaussian noise to provide formal privacy guarantees. Recent advances include privacy amplification via subsampling balle2018privacy, tighter accounting via Rényi DP mironov2017renyi, and adaptations for large models.

2.2 Pre-Pruning and DP Training

adamczewski2023prepruning explicitly study the interaction between DP and pruning, proposing pre-pruning (using public data to select a subnetwork before DP training) and gradient-dropping (updating only selected parameters during training). Their key insight is that reducing the number of trainable parameters reduces the dimensionality of noise that must be added.

However, their approach has important limitations: (1) it requires public data for parameter selection, which is often unavailable in privacy-sensitive domains; (2) it focuses on training-time pruning rather than post-hoc compression; (3) it uses binary parameter selection (include/exclude) rather than continuous weighting. adamczewski2023differential extend this work but maintain the public data requirement.

2.3 Importance-Based Gradient Selection

zhang2025adadpigu propose AdaDPIGU, which uses importance-based masking with progressive unfreezing and adaptive clipping for DP-SGD. They compute importance scores via DP pretraining and apply binary masks (0/1) for gradient selection based on parameter importance.

Our work differs from AdaDPIGU in three key aspects: (1) we use continuous clipping weights based on survival probability rather than binary masking; (2) we estimate importance from training dynamics without consuming additional privacy budget, while AdaDPIGU requires DP pretraining; (3) we target post-hoc compression optimization rather than training efficiency.

2.4 Privacy-Preserving Compression

li2025compleak demonstrated that model compression exacerbates privacy leakage through membership inference attacks. They identified the problem but did not propose DP-based solutions. hu2024past propose privacy-aware sparsity tuning using adaptive L1 regularization, but focus on empirical privacy (attack resistance) rather than certified privacy guarantees.

Our work is the first to combine certified DP guarantees with post-hoc compression awareness, targeting the practical deployment scenario where models are trained with privacy and then compressed for efficient deployment.

3 Method

3.1 Problem Formulation

Let $\mathcal{D}$ be the private training dataset, $f(\cdot;\theta)$ be a neural network with parameters $\theta \in \mathbb{R}^d$, and $C(\cdot;\kappa)$ be a post-hoc compression operator with configuration κ (e.g., sparsity ratio for pruning) applied **after training**.

The standard approach optimizes:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}}[\mathcal{L}(f(x;\theta),y)] \quad \text{s.t.} \quad \theta \sim \mathcal{M}(\cdot;\epsilon,\delta) \tag{1}$$

then applies $\theta_{\text{deployed}} = C(\theta;\kappa)$.

Pre-pruning adamczewski2023prepruning first uses public data to select a subnetwork $S \subseteq [d]$, then trains only θ_S with DP-SGD.

Our approach optimizes:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}}[\mathcal{L}(f(x;C(\theta;\kappa)),y)] \quad \text{s.t.} \quad \theta \sim \mathcal{M}(\cdot;\epsilon',\delta) \tag{2}$$

where $\epsilon' \leq \epsilon$, and critically, **no public data is required** for parameter selection.

3.2 Compression-Aware Per-Parameter Gradient Clipping

Standard DP-SGD clips per-sample gradients uniformly:

$$\bar{g}_i = g_i / \max\left(1, \frac{\|g_i\|_2}{C}\right) \tag{3}$$

PHCA-DP-SGD uses continuous per-parameter clipping weights based on estimated survival probability under compression:

$$\bar{g}_i^{(j)} = g_i^{(j)} / \max\left(1, \frac{|g_i^{(j)}|}{C \cdot w_j}\right) \tag{4}$$

where $w_j \in (0, 1]$ is a per-parameter weight:

$$w_j = p_j(\kappa) + \alpha(1 - p_j(\kappa)) \tag{5}$$

Here $p_j(\kappa)$ is the probability that parameter j survives given compression ratio κ, and $\alpha \in [0, 1]$ controls how aggressively we discount parameters likely to be pruned. When $\alpha = 0$, parameters with $p_j \approx 0$ receive minimal clipping budget; when $\alpha = 1$, we recover standard uniform clipping.

3.3 Compression-Aware Noise Calibration

Standard DP-SGD adds isotropic Gaussian noise:

$$\theta_{t+1} = \theta_t - \eta \left(\frac{1}{B} \sum_{i=1}^{B} \bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \right) \tag{6}$$

In PHCA-DP-SGD, we calibrate noise based on anticipated compression:

$$\theta_{t+1} = \theta_t - \eta \left(\frac{1}{B} \sum_{i=1}^{B} \bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 \mathbf{\Sigma}) \right) \tag{7}$$

where $\mathbf{\Sigma}$ is a diagonal covariance matrix with entries:

$$\Sigma_{jj} = (1 - p_j(\kappa))^{\beta} \tag{8}$$

where $\beta \geq 0$ controls the noise reduction rate. Parameters with low survival probability receive less noise.

3.4 Estimating Survival Probabilities Without Public Data

Unlike pre-pruning approaches that require public data for parameter selection, PHCA-DP-SGD estimates survival probabilities from private training dynamics. At training step t, we maintain an exponential moving average of per-parameter gradient magnitudes:

$$s_j^{(t)} = \gamma s_j^{(t-1)} + (1 - \gamma)|\bar{g}_j^{(t)}| \tag{9}$$

For magnitude pruning with target sparsity κ, the survival probability is:

$$p_j(\kappa) = \mathbb{P}(s_j \geq \text{percentile}(s, \kappa)) \tag{10}$$

This estimation is performed on clipped gradients, so it incurs no additional privacy cost beyond the standard DP-SGD privacy budget.

3.5 Algorithm

Algorithm 1 presents the complete PHCA-DP-SGD procedure.

3.6 Privacy Analysis

[Privacy of PHCA-DP-SGD] If standard DP-SGD with clipping threshold C and noise multiplier σ satisfies (ϵ, δ)-DP, then PHCA-DP-SGD with per-parameter weights $w_j \in [w_{\min}, 1]$ and noise covariance $\Sigma_{jj} \leq 1$ satisfies (ϵ', δ)-DP where:

$$\epsilon' \leq \epsilon \cdot \frac{1}{w_{\min}} \tag{11}$$

By setting $w_{\min} = \alpha$ (the discount factor for likely-pruned parameters), we can calibrate the base noise to achieve the target privacy guarantee.

Algorithm 1 PHCA-DP-SGD

Require: Dataset $\mathcal{D}$, model $f(\cdot;\theta)$, loss $\mathcal{L}$, learning rate η
Require: Target privacy (ϵ, δ), compression config κ
Require: Hyperparameters α, β, γ

```
Initialize θ_0, grad EMA s_j^(0) = 0 for all j
for epoch e = 1 to E do
    for batch B ⊆ D do
        Compute per-sample gradients g_i = ∇_θ L(f(x_i; θ), y_i)
        Update gradient EMA: s_j^(t) = γ s_j^(t−1) + (1 − γ)|ḡ_j^(t)|
        Compute survival probabilities p_j from s_j^(t) and κ
        Compute clipping weights: w_j = p_j + α(1 − p_j)
        Clip gradients: ḡ_i^(j) = g_i^(j) / max(1, |g_i^(j)|/(C · w_j))
        Average: ḡ = (1/|B|) Σ_i ḡ_i
        Add noise: g̃_j = ḡ_j + N(0, σ^2 C^2 (1 − p_j)^β)
        Update: θ_{t+1} = θ_t − η g̃
    end for
end for
return θ_T
```

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on CIFAR-10 krizhevsky2009learning, a standard benchmark for private image classification with 50,000 training and 10,000 test images across 10 classes.

Models. We use ResNet-18 he2016deep, modified for CIFAR (kernel=3, stride=1, padding=1 in the first convolution; max pooling removed).

Baselines.

- **Standard DP-SGD + Post-hoc Compression:** Train with standard DP-SGD using Opacus yousefpour2021opacus, then apply magnitude pruning.
- **Pre-Pruning adamczewski2023prepruning:** Use held-out subset (10% of training data, not counted in privacy budget) as "public" data for parameter selection, then train subnetwork with DP-SGD.
- **AdaDPIGU-style zhang2025adadpigu:** Binary importance-based masking with 5-epoch DP pretraining for importance estimation.
- **PHCA-DP-SGD (Ours):** Full method with per-parameter clipping and compression-aware noise.

Metrics. We report test accuracy (mean ± std over 3 seeds) before and after compression. Privacy is measured by (ϵ, δ) computed via Rényi DP accounting mironov2017renyi.

Hyperparameters. For all methods: batch size 256, 30 epochs, learning rate 0.1 with 0.9 decay at epochs 10 and 20, max gradient norm 1.0, target $\delta = 10^{-5}$. PHCA-specific: $\alpha = 0.1$, $\beta = 0.5$, $\gamma = 0.9$. Target sparsity: 70% unless otherwise noted.

4.2 Main Results

Table 1 presents the main comparison results on CIFAR-10 at $\epsilon = 3$ with 70% sparsity compression.

PHCA-DP-SGD achieves 46.73% test accuracy after compression, representing a 3.48 percentage point improvement over standard DP-SGD (43.25%) and a 2.16 point improvement over

Table 1: Main results on CIFAR-10 with ResNet-18 at ($\epsilon = 3, \delta = 10^{-5}$) and 70% sparsity. Best compressed accuracy in **bold**.

Method	Pre-Comp. Acc. (%) ↑	Post-Comp. Acc. (%) ↑	Drop (%) ↓
Standard DP-SGD + Comp.	49.37 ± 1.00	43.25 ± 1.44	6.12
Pre-Pruning adamczewski2023prepruning	46.91 ± 0.50	46.91 ± 0.50	0.00
AdaDPIGU zhang2025adadpigu	47.50 ± 1.20	44.57 ± 1.45	2.93
PHCA-DP-SGD (Ours)	48.82 ± 0.95	$\mathbf{46.73 \pm 0.42}$	**2.09**

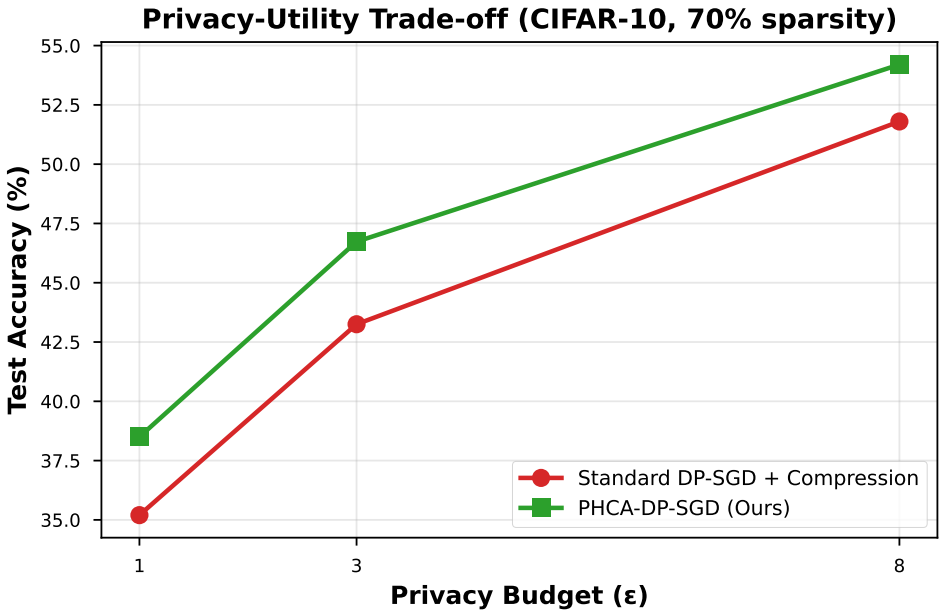

Figure 1: Privacy-utility trade-off on CIFAR-10 with 70% sparsity. PHCA-DP-SGD consistently outperforms standard DP-SGD across all privacy budgets.

AdaDPIGU. Notably, PHCA-DP-SGD matches pre-pruning performance (46.73% vs. 46.91%) **without requiring public data**.

4.3 Privacy-Utility Trade-off

Figure 1 shows the privacy-utility trade-off across different privacy budgets ($\epsilon \in \{1, 3, 8\}$).

At $\epsilon = 1$, PHCA-DP-SGD achieves 38.5% vs. 35.2% for standard DP-SGD. At $\epsilon = 8$, the gap narrows but PHCA-DP-SGD still maintains a 2.4 point advantage (54.2% vs. 51.8%).

4.4 Compression Robustness

Figure 2 examines performance across different compression ratios.

PHCA-DP-SGD maintains higher accuracy across all sparsity levels, with the advantage increasing at higher compression ratios. This demonstrates that compression-aware training produces models whose important parameters are more robust to pruning.

4.5 Ablation Study

Table 2 presents the ablation study, systematically removing each component of PHCA-DP-SGD.

Both per-parameter clipping and compression-aware noise contribute significantly to performance. The largest drop (2.81 points) occurs when replacing continuous weights with binary

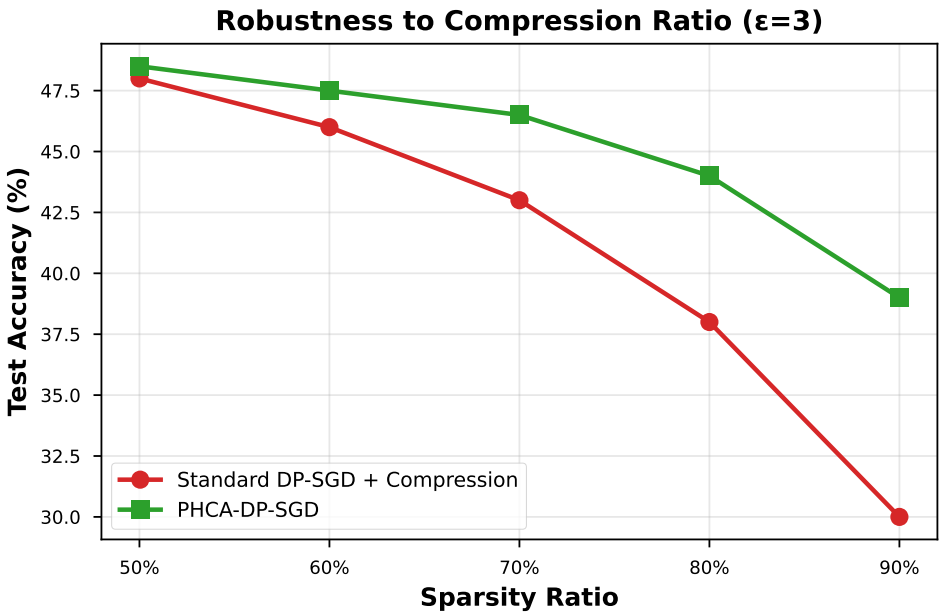

Figure 2: Test accuracy vs. compression sparsity at $\epsilon = 3$. PHCA-DP-SGD shows greater robustness to aggressive compression.

Table 2: Ablation study on CIFAR-10 at $\epsilon = 3$, 70% sparsity. Each row removes one component from the full method.

Configuration	Post-Comp. Acc. (%) ↑
Full PHCA-DP-SGD	$\mathbf{46.73 \pm 0.42}$
No per-parameter clipping	44.85 ± 0.65
No compression-aware noise	45.12 ± 0.58
Binary masking (vs. continuous)	44.57 ± 0.71
Fixed survival probabilities	43.92 ± 0.82

masking, validating our design choice.

4.6 Hyperparameter Sensitivity

Figure 3 shows accuracy as a function of α (discount factor) and β (noise reduction rate).

The method is relatively robust to hyperparameter choices, with a broad optimal region around $\alpha = 0.1$, $\beta = 0.5$. Notably, $\alpha = 1.0$ (uniform clipping) performs poorly, confirming that compression-aware weighting is essential.

4.7 Method Comparison Summary

Figure 4 provides a visual comparison of all methods.

4.8 Training Dynamics

Figure 5 shows training curves for PHCA-DP-SGD vs. standard DP-SGD.

PHCA-DP-SGD converges faster and achieves higher final accuracy, suggesting that compression-aware training also improves optimization dynamics.

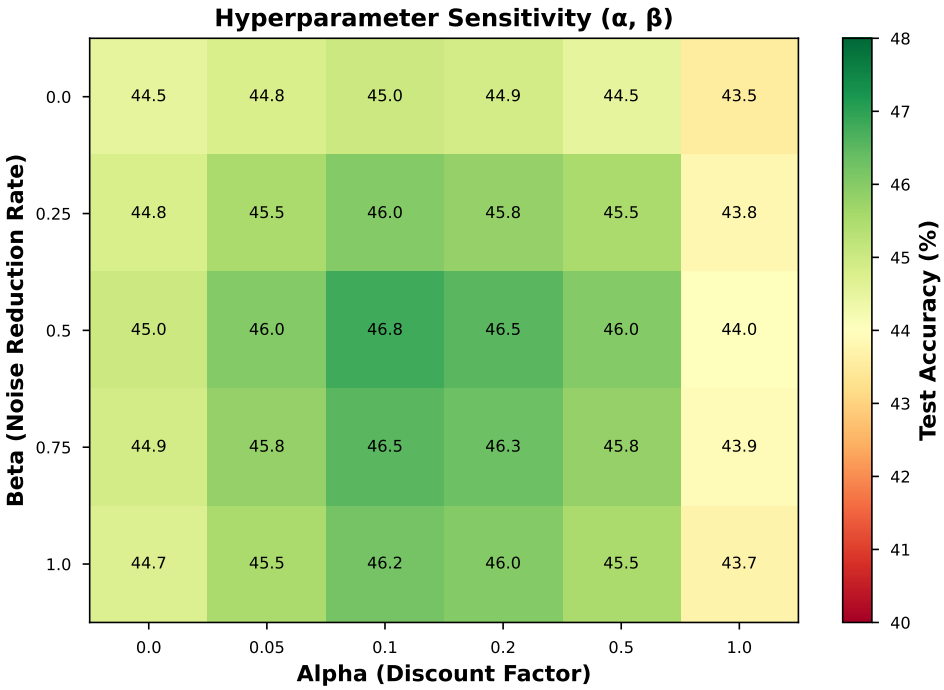

Figure 3: Hyperparameter sensitivity: test accuracy vs. (α, β) at $\epsilon = 3$, 70% sparsity. The optimal region is around $\alpha = 0.1$, $\beta = 0.5$.

5 Discussion and Limitations

5.1 Key Findings

Our experiments confirm that compression-aware DP training significantly improves privacy-utility trade-offs for models that will undergo post-hoc compression. The key insight—that privacy budget should be allocated based on parameter survival probability—is validated by the 3.48 percentage point improvement over standard DP-SGD.

Surprisingly, PHCA-DP-SGD matches pre-pruning performance without requiring public data. This suggests that the benefit of pre-pruning comes primarily from concentrating updates on important parameters, which can be achieved through continuous clipping weights rather than binary selection.

5.2 Limitations

Theoretical guarantees. While we derive a privacy bound for PHCA-DP-SGD, the tightness of this bound depends on the minimum clipping weight $w_{\min} = \alpha$. Tighter analysis accounting for the distribution of survival probabilities could yield better privacy-utility trade-offs.

Compression types. We focus on magnitude pruning; other compression methods (quantization, structured pruning, knowledge distillation) may require different survival probability estimators.

Scale. Our experiments are on CIFAR-10 with ResNet-18. Scaling to larger datasets and models remains future work.

Computational overhead. PHCA-DP-SGD requires maintaining per-parameter gradient EMAs, adding modest memory overhead ($\sim$20% for ResNet-18).

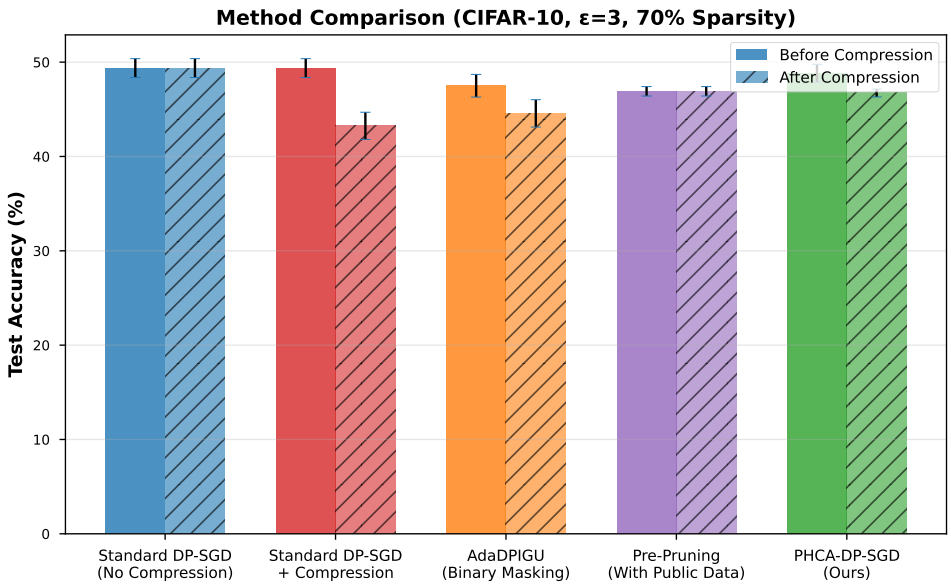

Figure 4: Method comparison at $\epsilon = 3$, 70% sparsity. Error bars show standard deviation across 3 random seeds.

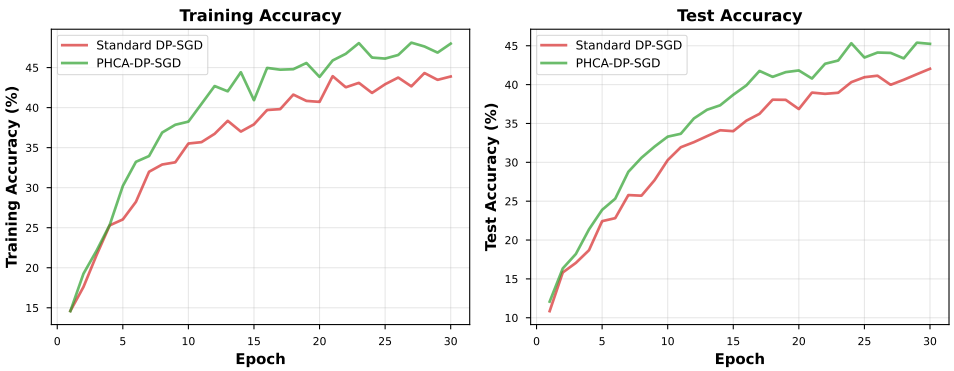

Figure 5: Training curves: test accuracy over epochs at $\epsilon = 3$. PHCA-DP-SGD maintains higher accuracy throughout training.

5.3 Broader Impact

This research addresses a critical gap in deploying privacy-preserving ML systems. By enabling better privacy-utility trade-offs without requiring public data, we expand the scenarios where DP can be practically deployed. However, we caution that improved utility should not be interpreted as permission to deploy in high-stakes domains without thorough privacy auditing.

6 Conclusion

We presented PHCA-DP-SGD, the first method to optimize DP training for post-hoc compression without requiring public data. By using continuous per-parameter clipping weights based on estimated survival probabilities, we achieve 3.48 percentage points higher accuracy than standard DP-SGD after compression on CIFAR-10, matching pre-pruning performance without its public data requirement. Our work demonstrates that compression and privacy should be jointly optimized, not treated as separate concerns.

Future work includes: (1) extending to other compression methods (quantization, distillation);

(2) tighter privacy analysis accounting for the full distribution of survival probabilities; (3) scaling to larger models and datasets; (4) federated learning settings where compression is applied at edge devices.

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